Grammatical Gender in German Nouns: Article Distribution and Ending Patterns

Nov 8, 2026 · @Carlos

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- **Third-party material:** the Duden and Arabi (2017) figures are cited for comparison only and remain the property of their respective owners.
- **Scope:** results describe a random sample of 10,000 German nouns, not the German language as a whole, and have not been peer reviewed.
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Executive Summary

A random sample of 10,000 German nouns reproduces the gender structure published by Duden: feminine nouns are the largest group, masculine the second and neuter the smallest. The sample's article shares sit within 2.4 percentage points of Duden's, and three suffix rules checked against Arabi (2017) land within about 4 points of his rates.

- **Article distribution:** die 43.59% (4,359 nouns), der 35.96% (3,596), das 20.45% (2,045).
- **Ending coverage:** 47.77% of nouns (4,777) end in one of 21 common endings: 68.48% of die nouns, 35.62% of der and 24.99% of das. The ending -e drives much of the feminine coverage; without it, total coverage falls to 29.12%.

- **Most useful single rules:** -e points to die in 88.47% of cases, -er to der in 78.20% and -um to das in 71.43%.
- **Near-certain rules:** -ung, -heit, -keit, -schaft, -tät and -ion point to die in 97.98% to 100% of their nouns.
- **Hardest article:** neuter has the lowest coverage, and no catalog ending covers more than 8.31% of its nouns (-er).

Differences from Duden of up to 2.4 points exceed the sampling margin of roughly ±1 point, so differences between the word lists likely contribute alongside sample size (see Benchmark Comparison).

Data and Method

The analysis uses a reproducible random sample of 10,000 unique noun–article pairs, drawn from 84,281 filtered pairs of the german-nouns dataset (11.87% of the filtered list). The filters and the data source are listed in the References.

- **Article:** the definite article of the nominative singular (m = der, f = die, n = das). Nouns listed with more than one gender were removed, so every noun has exactly one article.
- **Validation:** the file has 10,000 rows, no blank fields, no duplicate words, and every word starts with a capital letter. The article counts (4,359 die, 3,596 der, 2,045 das) were reproduced independently in PostgreSQL and Power BI.
- **Ending catalog:** 21 endings, grouped by the article they are traditionally associated with. Feminine: -ung, -heit, -keit, -schaft, -tät, -ion, -sion, -ik, -e. Masculine: -er, -or, -ling, -ismus, -ig, -ant, -ist. Neuter: -chen, -lein, -ment, -um, -ma.
- **Matching rule:** case-insensitive, and the word must be longer than the ending. A noun that matches several endings (for example -ion and -sion, 18 nouns) counts once in coverage figures and under each matching ending in ending-level rates.
- **Share of article:** nouns with that article divided by the 10,000 nouns in the sample.
- **Coverage:** nouns of an article with at least one catalog ending, divided by all nouns of that article.
- **Ending reliability:** among the nouns with a given ending, the share that carry the article the ending is most often associated with in the sample.
- **Uncertainty:** 95% Wilson intervals. For article shares the margin is about ±1 percentage point; it is wider for rare endings.

Results

Feminine nouns make up 43.59% of the sample, masculine 35.96% and neuter 20.45%, and 47.77% of all nouns end in at least one catalog ending.

Table 1. Article distribution and ending coverage (n = 10,000)

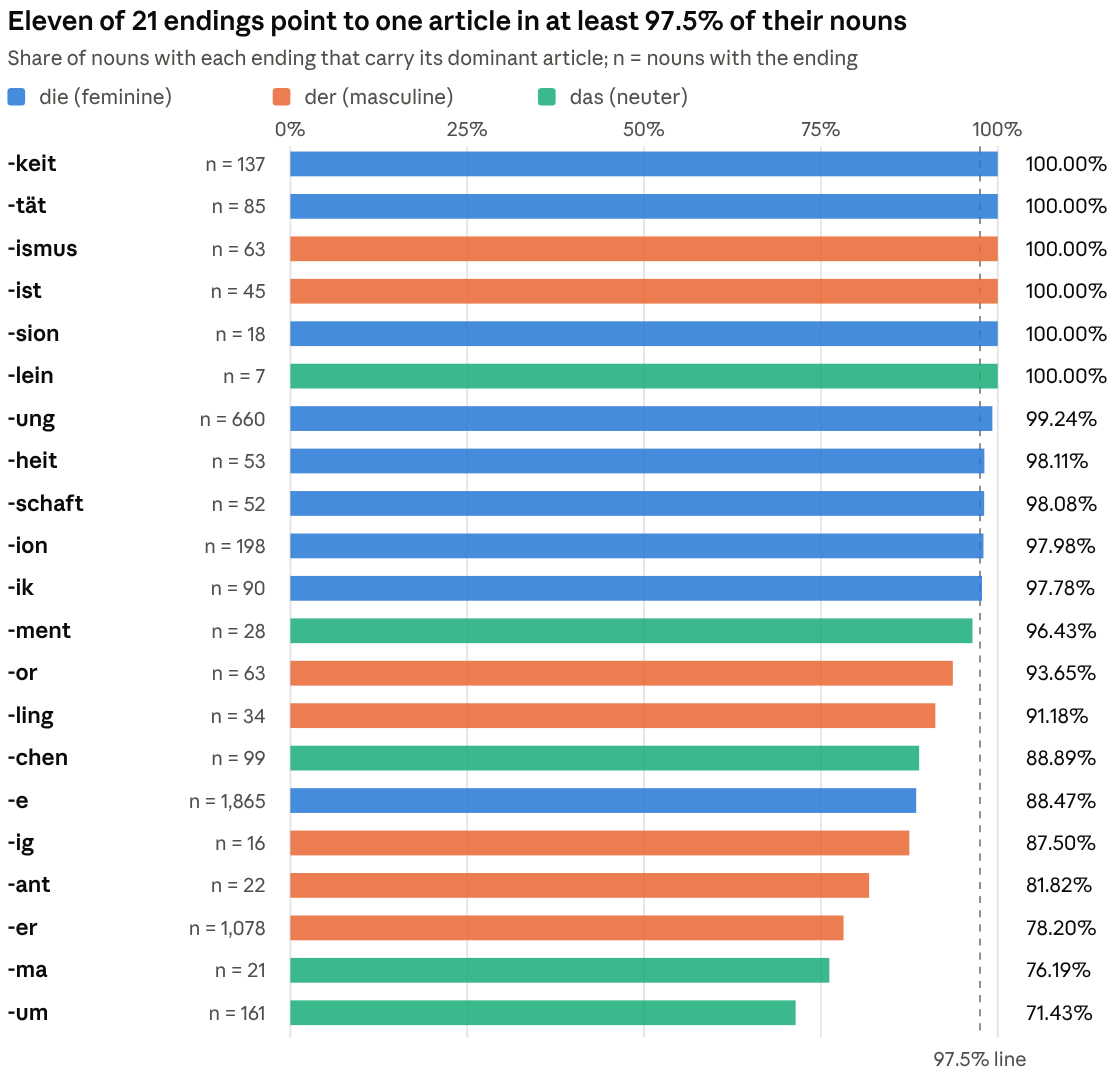
Article	Nouns	Share of sample	95% interval	With a catalog ending	Coverage	Most common ending
die (feminine)	4,359	43.59%	42.62–44.56%	2,985	68.48%	-e (37.85% of die nouns)
der (masculine)	3,596	35.96%	35.02–36.91%	1,281	35.62%	-er (23.44% of der nouns)
das (neuter)	2,045	20.45%	19.67–21.25%	511	24.99%	-er (8.31% of das nouns)
All nouns	10,000	100%	—	4,777	47.77%	—

Coverage depends on -e. Without it, coverage falls from 68.48% to 30.63% for die, from 35.62% to 31.65% for der and from 24.99% to 21.47% for das (47.77% to 29.12% overall). The feminine advantage rests on one very frequent ending: 1,865 nouns (18.65% of the sample) end in -e, and 1,650 of them are feminine.

Neuter is the hardest article to characterize. Within the catalog its most frequent endings are -er (8.31% of das nouns), -um (5.62%) and -chen (4.30%). Outside the catalog, -en is more frequent than any of them: 236 neuter nouns (11.54%) end in it.

Reliability of endings

Eleven of the 21 endings point to a single article in at least 97.5% of their nouns; -um, -ma and -er are the least decisive.



computed from the 10,000-noun sample · 21 catalog endings

Each bar shows the share of nouns with that ending that carry its dominant article. Counts are small for some endings (for example -lein, n = 7), so their rates are less certain than those of -ung, -keit or -e.

Benchmark Comparison

The sample reproduces Duden's ranking of the three articles and the direction of all three Arabi rules, staying within 2.4 percentage points of Duden's shares and about 4 points of Arabi's rates.

Article distribution against Duden

Table 2. Share of nouns by article: sample against Duden

Article	This sample	95% interval	Duden	Difference (points)
die (feminine)	43.59%	42.62–44.56%	45.2%	−1.61
der (masculine)	35.96%	35.02–36.91%	33.6%	+2.36
das (neuter)	20.45%	19.67–21.25%	21.2%	−0.75

Duden's figures cover the 98.4% of nouns in its spelling dictionary that take a single article. It states that 45.2% of them are feminine, 11.6 points more than masculine (33.6%); the neuter share (21.2%) is the remainder. The sample uses the same basis, because nouns with several articles were removed.

The neuter gap lies inside the sample's interval, but the feminine and masculine gaps do not: they are larger than the sampling margin of about ±1 point. Sample size therefore explains part of the difference, and the rest most likely comes from the word lists. This sample draws on a Wiktionary-derived list of single-word common nouns, while Duden counts the single-article nouns of its own spelling dictionary.

Ending reliability against Arabi (2017)

Table 3. Share of nouns with each ending that carry its expected article: sample against Arabi (2017)

Ending	Expected article	This sample	Nouns with the ending	95% interval	Arabi (2017)	Difference (points)
-er	der	78.20%	1,078	75.64–80.56%	79.69%	−1.49
-e	die	88.47%	1,865	86.94–89.84%	91.78%	−3.31
-um	das	71.43%	161	64.01–77.84%	67.41%	+4.02

The three rules point to the same article in both studies. Arabi's rates for -er and -um fall inside the sample's 95% interval. His rate for -e is slightly above it (upper bound 89.84%), which is consistent with the two studies using different word lists.

Limitations

The results describe this sample and these 21 endings, and the comparisons with Duden and Arabi are indicative rather than exact.

- **Sample size:** 10,000 of 84,281 filtered pairs (11.87%). Article shares carry a margin of about ±1 point; rare endings carry far more (-lein has only 7 nouns).
- **Word list:** the list comes from German Wiktionary and has no usage frequency, so it mixes common and rare words. Shares describe vocabulary types, not how often each article appears in speech or writing. Compounds, which take the gender of their final element, are common in such a list.
- **Single gender only:** nouns listed with several genders (417 words in the source filtering) were removed, so the analysis says nothing about them.
- **Catalog:** the 21 endings were chosen by the author and are not exhaustive. For example, -en is frequent among neuter nouns (236 nouns) but is not in the catalog.
- **Spelling, not morphology:** endings are matched as letter strings. The -er in Mutter is not an agent suffix and the -e in Name is not a feminine marker, so ending rates describe spelling patterns.
- **Different sources:** Duden and Arabi use different dictionaries, dates and methods, and Arabi's figures were not re-derived here.

Conclusions

The sample reproduces Duden's gender ranking and the direction of Arabi's suffix rules, which supports three conclusions.

1. **The gender structure holds in a sample:** die 43.59%, der 35.96% and das 20.45% follow Duden's order (45.2%, 33.6%, 21.2%), with differences of 0.75 to 2.36 points.
2. **Endings are strong but uneven predictors:** -ung, -heit, -keit, -schaft, -tät and -ion are near-certain for die, while -e (88.47%), -er (78.20%) and -um (71.43%) are useful but leave room for exceptions.
3. **Neuter needs more than endings:** it has the lowest coverage (24.99%), and no catalog ending covers more than 8.31% of its nouns.

A larger sample, the full 84,281-pair list and a data-driven ending catalog (for example, adding -en) would narrow the gaps and test the neuter pattern.

References

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Data source and method

Table 4. Source of the word list and how the article was obtained

Item	Description
Data source	german-nouns (gambolputty), a list of German nouns with their grammatical gender
Original source	German Wiktionary (WiktionaryDE), compiled from the official dump
Repository	https://github.com/gambolputty/german-nouns (file german_nouns/nouns.csv)
License	CC BY-SA 4.0 (Creative Commons Attribution-ShareAlike 4.0). Attribution: Wiktionary contributors and gambolputty.
How the article was obtained	Gender in the source: m = der, f = die, n = das (definite article, nominative singular).
Note	The source has no usage frequency, so the sample mixes common and uncommon words.

Table 5. Filters applied, from the original file to the sample

Step	Filter	Rows remaining
0	Rows in the original CSV	102,444
1	Common nouns only (type 'Substantiv'). Place names, proper names, surnames, affixes, abbreviations and multi-word expressions are excluded.	90,612
2	Only nouns with a single gender (m/f/n)	87,788
3	Only nouns with a nominative singular form	87,710
4	Single word, letters only, initial capital	85,506
5	417 words listed with different genders (for example 'der/die See') are discarded to avoid ambiguous articles	84,281 unique pairs
6	Reproducible random sample (seed 2026), sorted alphabetically	10,000 pairs